

Laminar flow theory
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Laminar flow theory 2021.09.15

Text book..

F. W. White, Viscous fluid flow,
McGraw Hill.

References:

1. H. Schlichting, Boundary layer theory
McGraw Hill.
2. G. K. Batchelor, An introduction
to fluid mechanics.
3. D. J. Tritton, Physical fluid dynamics
Oxford Science Publication.
4. R. L. Panton, Incompressible flow,
Wiley.

Contents:

1. Preliminary concept.
2. Fundamental equations of viscous flow.
3. Solutions of Newtonian viscous flows.
4. Laminar boundary layer
— similarity solution

Grading policy:

mid term exam

final exam

30% ← November

30% ← January

Take home assignments

1. flow over a cylinder 15% October.



$$Re = \frac{\rho U D}{\mu}$$

D : diameter of cylinder

• At low Re , the flow is steady.

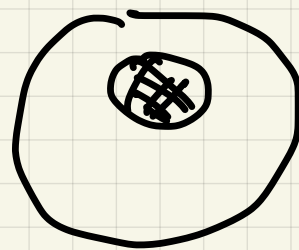
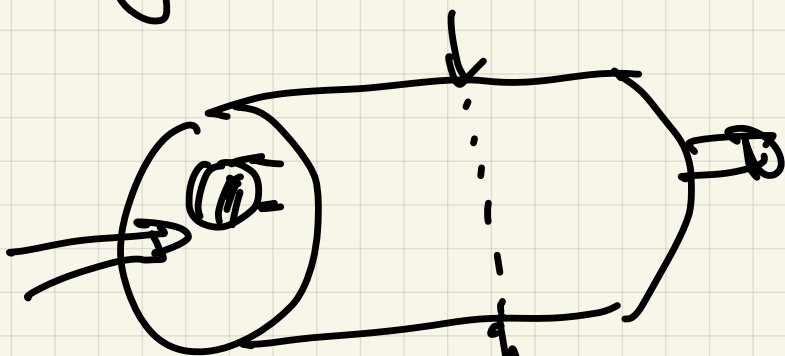
• At higher Reynolds number, the flow is unsteady



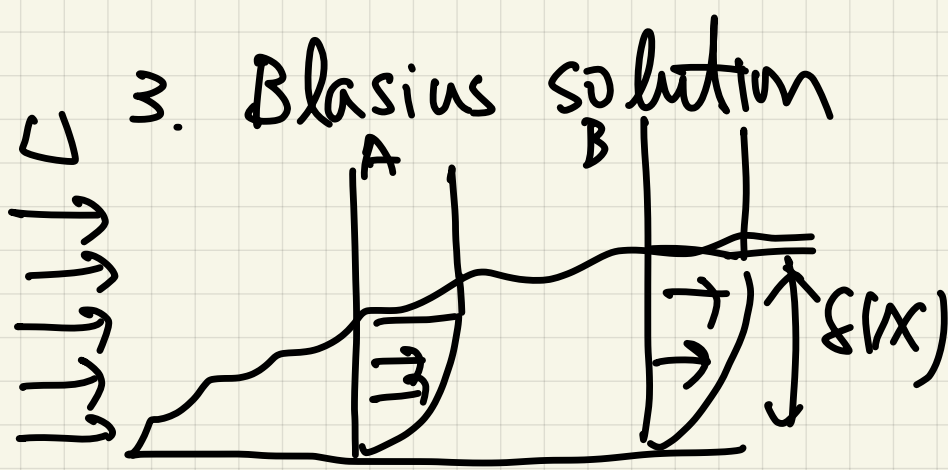
- use ANSYS to simulate the flow field.

2. flow within a complex geometry 10%

November

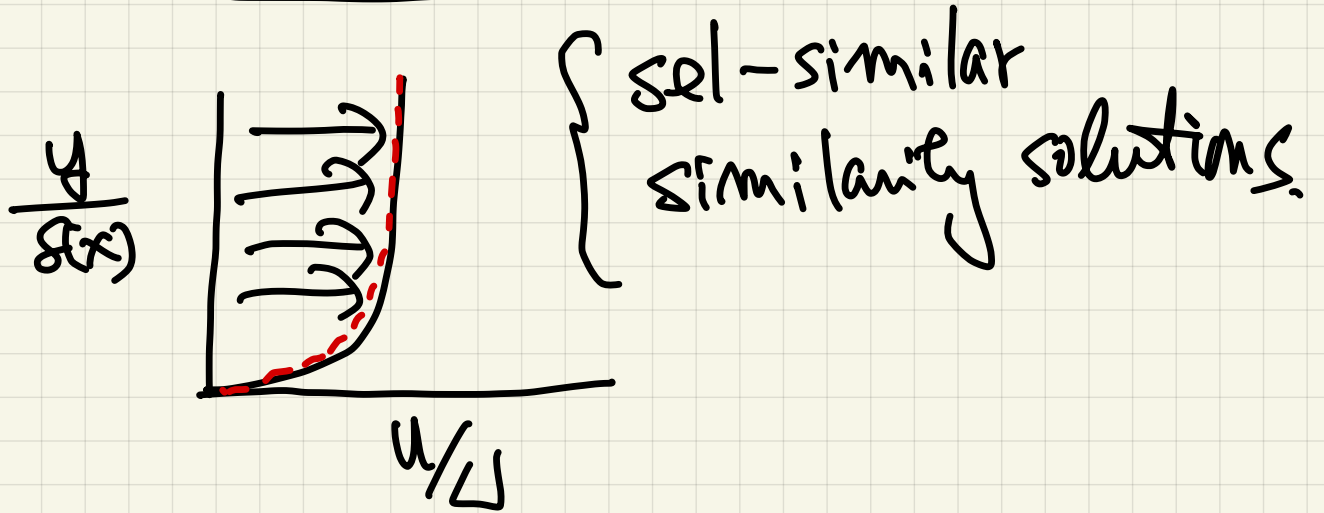


simulation $\Leftrightarrow$ analytic solutions



15%

December.



— All the take home assignments must be submitted to the turnitin website.

equations:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \rho \vec{v} = 0 \quad - \text{continuity}$$

$$\frac{\partial \rho \vec{v}}{\partial t} + \nabla \cdot \rho \vec{v} \vec{v} = -\nabla p + \nabla \cdot \left[\mu (\nabla \vec{v} + (\nabla \vec{v})^T) - \lambda \nabla \cdot \vec{v} \vec{I} \right] \quad - \text{N-S equation}$$

$$\frac{\partial p}{\partial t} + \frac{\partial}{\partial x}(\rho u) + \frac{\partial}{\partial y}(\rho v) + \frac{\partial}{\partial z}(\rho w) = 0.$$

u momentum. - Cartesian coordinate

$$\frac{\partial \rho u}{\partial t} + \frac{\partial}{\partial x}(\rho u u) + \frac{\partial}{\partial y}(\rho v u) + \frac{\partial}{\partial z}(\rho w u)$$

$$= \frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x} \left[\mu \left(\frac{\partial u}{\partial x} + \frac{\partial u}{\partial x} \right) - \lambda \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \right) \right]$$

$$+ \frac{\partial}{\partial y} \left[\mu \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) \right]$$

$$+ \frac{\partial}{\partial z} \left[\mu \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} \right) \right]$$

- Tensor notation

$$\vec{v} = u \vec{i} + v \vec{j} + w \vec{k} = u_i, \quad i=1,2,3$$

$$u_1 = u, \quad u_2 = v, \quad u_3 = w$$

$$\vec{\nabla} = \frac{\partial}{\partial x} \vec{i} + \frac{\partial}{\partial y} \vec{j} + \frac{\partial}{\partial z} \vec{k}$$

$$= \frac{\partial}{\partial x_i}, \quad i=1,2,3$$

$$\frac{\partial}{\partial x_1} = \frac{\partial}{\partial x}, \quad \frac{\partial}{\partial x_2} = \frac{\partial}{\partial y}, \quad \frac{\partial}{\partial x_3} = \frac{\partial}{\partial z}$$

$$\vec{\nabla} \cdot \vec{v} = \left(\frac{\partial}{\partial x} \vec{i} + \frac{\partial}{\partial y} \vec{j} + \frac{\partial}{\partial z} \vec{k} \right) \cdot (u \vec{i} + v \vec{j} + w \vec{k})$$

scalar

$$= \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = \frac{\partial u_i}{\partial x_i} = \frac{\partial u_j}{\partial x_j}$$

repeated indices.

$$\frac{\partial u_i}{\partial x_i} = \frac{\partial u_1}{\partial x_1} + \frac{\partial u_2}{\partial x_2} + \frac{\partial u_3}{\partial x_3}$$

$$\vec{\nabla} \vec{v} = \left(\frac{\partial}{\partial x} \vec{i} + \frac{\partial}{\partial y} \vec{j} + \frac{\partial}{\partial z} \vec{k} \right) (u \vec{i} + v \vec{j} + w \vec{k})$$

$$= \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{pmatrix} = \frac{\partial u_i}{\partial x_j}, \quad i=1,2,3, \quad j=1,2,3$$

2nd rank tensor

$$\nabla p = \frac{\partial p}{\partial x} \vec{e}_1 + \frac{\partial p}{\partial y} \vec{e}_2 + \frac{\partial p}{\partial z} \vec{e}_3$$

$$= \frac{\partial p}{\partial x_i}, \quad i=1,2,3$$

$$\nabla \cdot \nabla \vec{u} = \frac{\partial}{\partial x_i} \frac{\partial u_i}{\partial x_j} \leftarrow \text{vector}$$

$$= \frac{\partial^2 u_1}{\partial x_1^2} + \frac{\partial^2 u_1}{\partial x_2^2} + \frac{\partial^2 u_1}{\partial x_3^2}$$

$$\nabla \cdot (\nabla \vec{u})^T = \frac{\partial}{\partial x_j} \left(\frac{\partial u_i}{\partial x_j} \right)$$

$$= \frac{\partial}{\partial x_1} \left(\frac{\partial u_1}{\partial x_1} \right) + \frac{\partial}{\partial x_2} \left(\frac{\partial u_2}{\partial x_1} \right) + \frac{\partial}{\partial x_3} \left(\frac{\partial u_3}{\partial x_1} \right)$$

$$\vec{e}_1 = 1$$

$$\nabla \cdot [\nabla \vec{u} + (\nabla \vec{u})^T] = \frac{\partial^2 u_1}{\partial x_1^2} + \frac{\partial^2 u_1}{\partial x_2^2} + \frac{\partial^2 u_1}{\partial x_3^2}$$

$$+ \frac{\partial}{\partial x_1} \left(\frac{\partial u_1}{\partial x_1} \right) + \frac{\partial}{\partial x_2} \left(\frac{\partial u_2}{\partial x_1} \right) + \frac{\partial}{\partial x_3} \left(\frac{\partial u_3}{\partial x_1} \right)$$

$$= \frac{\partial}{\partial x_1} \left(\frac{\partial u_1}{\partial x_1} + \frac{\partial u_1}{\partial x_1} \right) + \frac{\partial}{\partial x_2} \left(\frac{\partial u_1}{\partial x_2} + \frac{\partial u_2}{\partial x_1} \right)$$

$$+ \frac{\partial}{\partial x_3} \left(\frac{\partial u_1}{\partial x_3} + \frac{\partial u_3}{\partial x_1} \right)$$

$$\frac{\partial p}{\partial t} + \nabla \cdot p \vec{v} = 0$$

$$\frac{\partial p}{\partial t} + \frac{d}{dt}(p u_i) = 0 \quad (A)$$

$$\frac{\partial p \vec{v}}{\partial t} + \nabla \cdot p \vec{v} \vec{v}^T = -\nabla p + \nabla \cdot \left[\mu (\nabla \vec{v} + (\nabla \vec{v})^T) - \lambda \nabla \cdot \vec{v} \vec{I} \right]$$

$$\frac{\partial p u_i}{\partial t} + \frac{d}{dt}(p u_j u_i) = -\frac{\partial p}{\partial x_i} + \frac{d}{dt} \left[\mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) - \lambda \frac{\partial u_k}{\partial x_k} \delta_{ij} \right] \quad (B)$$

$$\delta_{ij} : \begin{matrix} 1 & i=j \\ 0 & i \neq j \end{matrix}$$

Newtonian

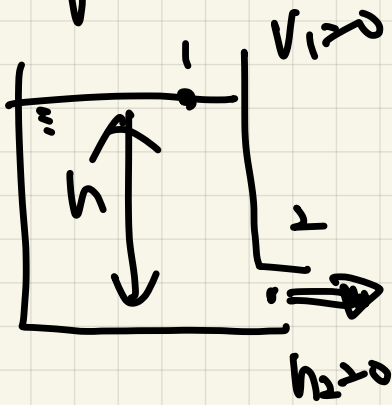
$$\tau = \mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \dots$$

- Take home.

expand equations (A) and (B) in Cartesian coordinates.

Chapter 1: Preliminary concept.

- fluid mechanics.



$$v = \sqrt{2gh} \quad \text{Torricelli (1608 ~ 1647)}$$

$$\frac{1}{2}v^2 + gh = C.$$

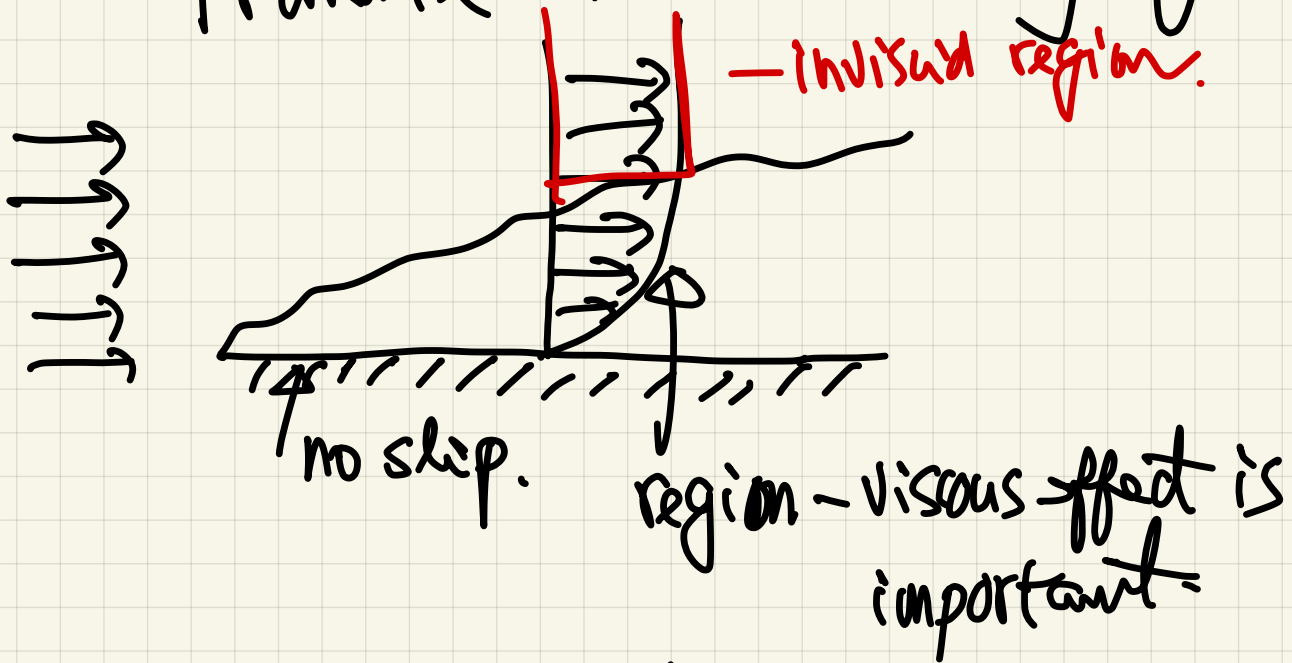
$$gh_1 = \frac{1}{2}v_2^2.$$

Euler equation. $\frac{dP u_i}{dt} + \frac{d}{dx_j} (P g_j u_i) = - \frac{dP}{dx_i}.$
Inviscid.

Navier (1827) Stokes (1845)

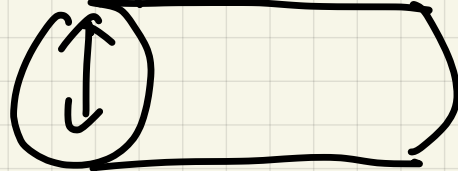
$$\frac{dP u_i}{dt} + \frac{d}{dx_j} (P g_j u_i) = - \frac{dP}{dx_i} + \frac{d}{dx_j} \left[\mu \left(\frac{du_i}{dx_j} + \frac{du_j}{dx_i} \right) - \lambda \nabla \cdot \vec{v} \delta_{ij} \right]$$

- Prandtl 1904 Boundary layer.

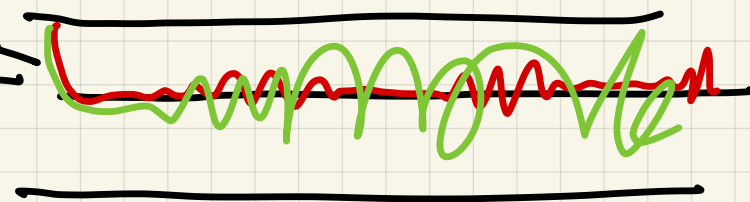


- Reynolds number D diameter

$$Re = \frac{\rho U D}{\mu}$$



dye



intermediate Re

low Re .

high Re . turbulent flow

$$Re = \frac{\rho U D}{\mu} \quad \frac{\text{dynamic force}}{\text{viscous force.}}$$

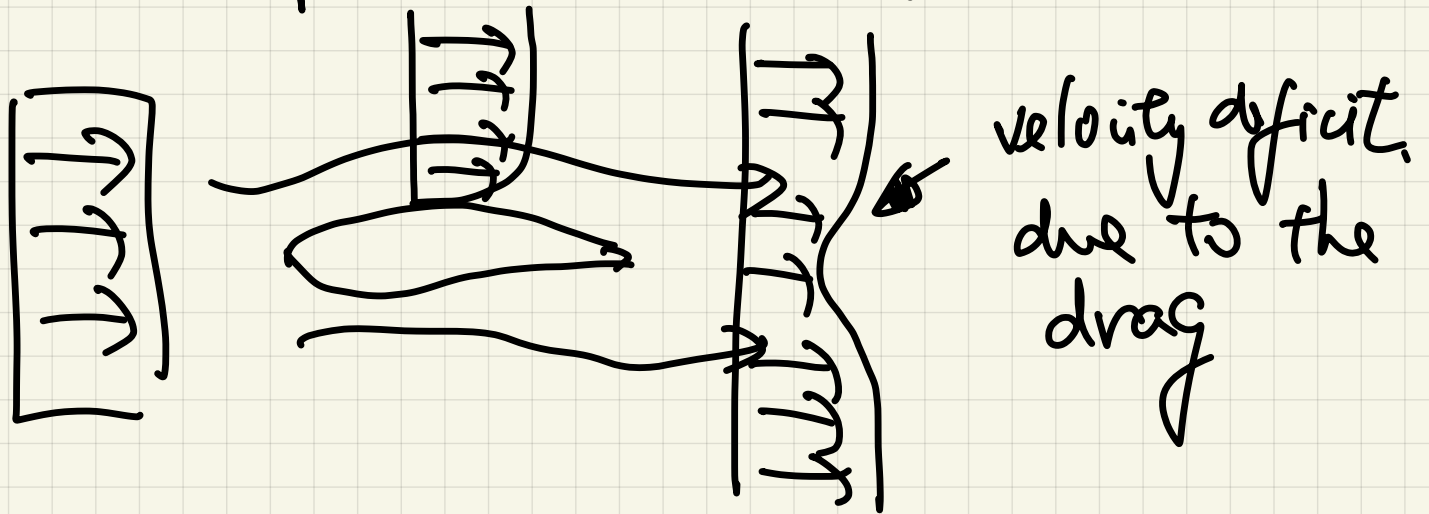
- Flow visualization, Van Dyke (1982)

- Flow simulation $\Leftrightarrow$ Computational fluid dynamics CFD.

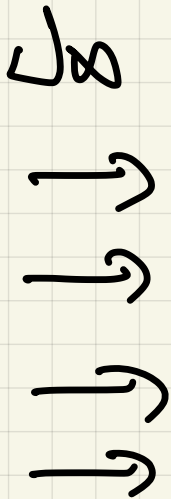
Examples:

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- Flow past a thin airfoil



- Flow past a circular cylinder

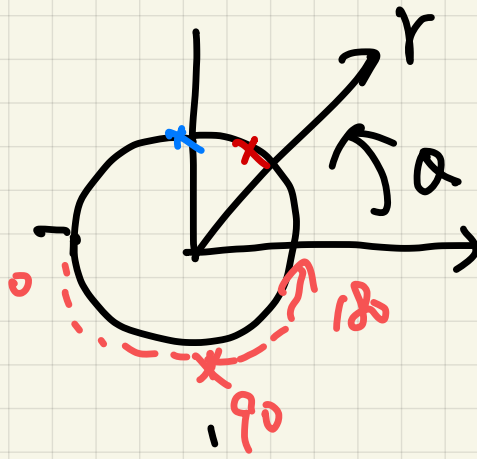


Inviscid flow

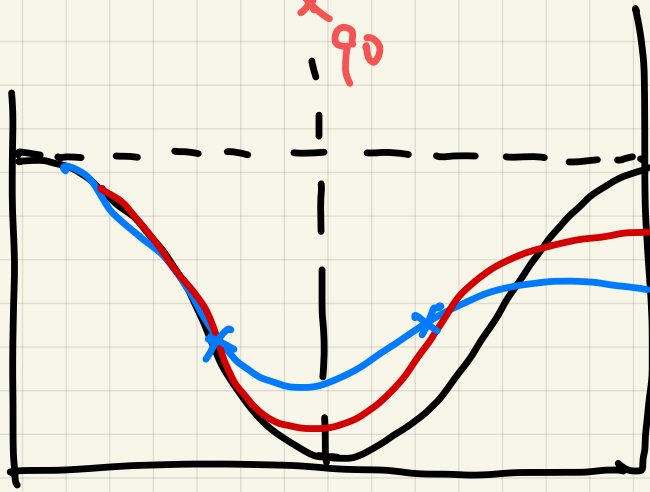
$$v_r = U_\infty \left(1 - \frac{R^2}{r^2}\right) \cos \theta$$

$$v_\theta = U_\infty \left(1 + \frac{R^2}{r^2}\right) \sin \theta$$

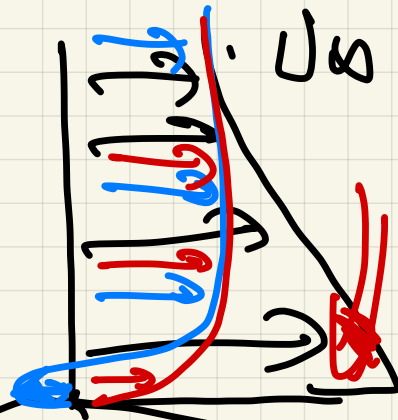
$$P_s = P_0 + \frac{1}{2} \rho U_\infty^2 (1 - 4 \sin^2 \theta)$$



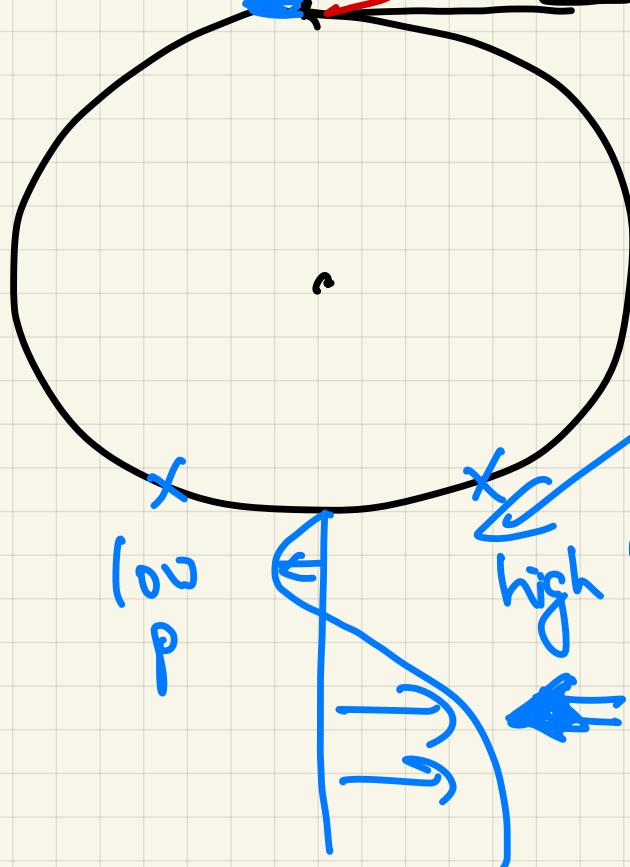
$$P_s = P_0 + \frac{1}{2} \rho U_\infty^2 (1 - \sin^2 \alpha)$$



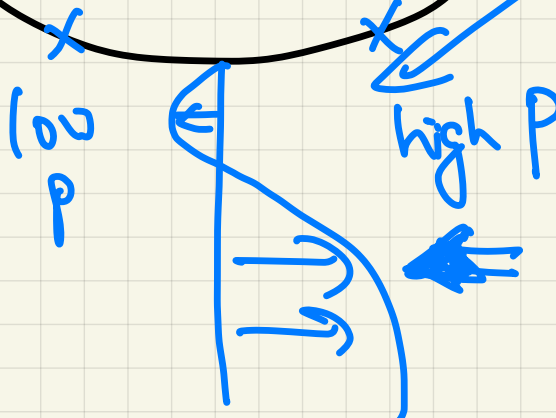
— inviscid.
 — : laminar
 — : turbulent



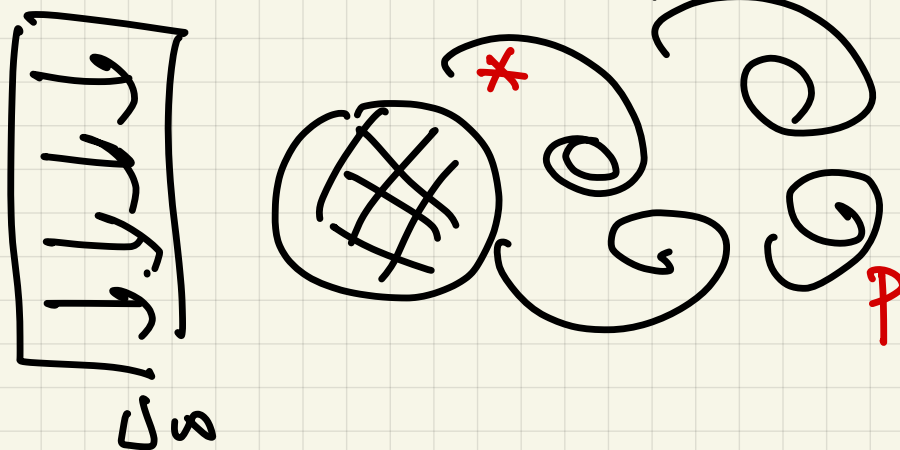
Turbulence promotes mixing.



adverse pressure gradient



- Vortex shedding



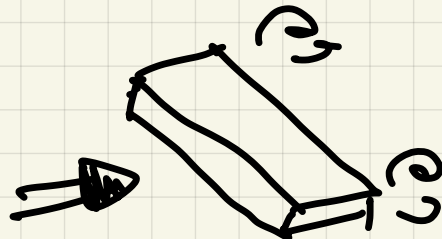
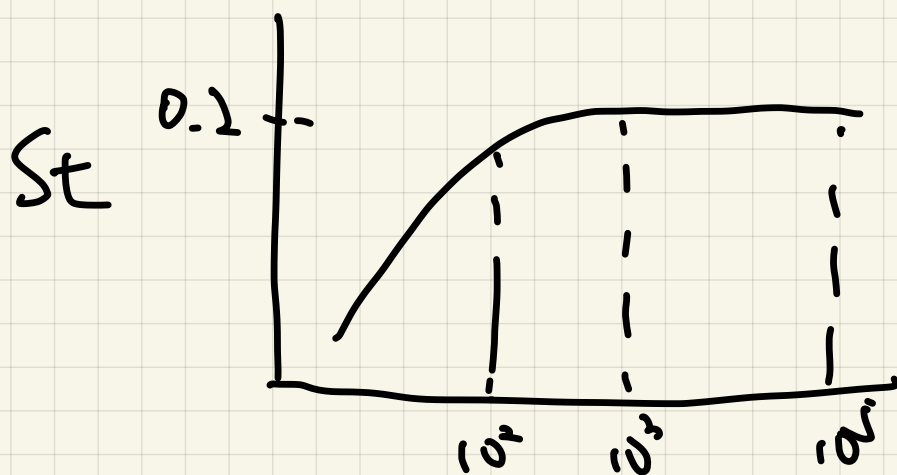
$$Re = \frac{\rho U D}{\mu} > 47.$$



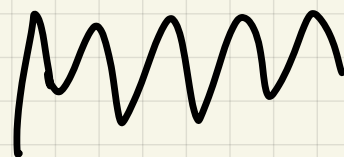
- Strouhal number

$$St = \frac{f D}{U_\infty} \approx 0.2.$$

$$Re = 100 \sim 10^5$$

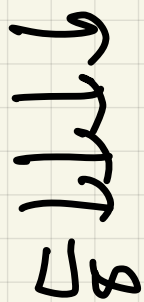


- singing wire



- drag coefficient.

$$C_D = \frac{\text{Drag}}{\rho U_\infty^2 D}$$



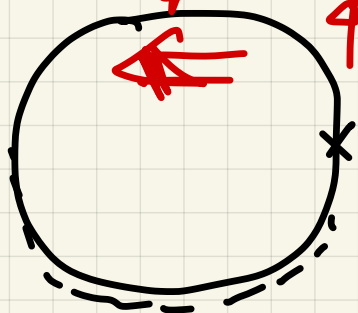
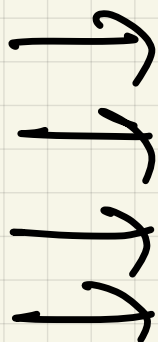
early transition
to turbulence
at lower Re

transition to
turbulence.

C_D

surface
roughness

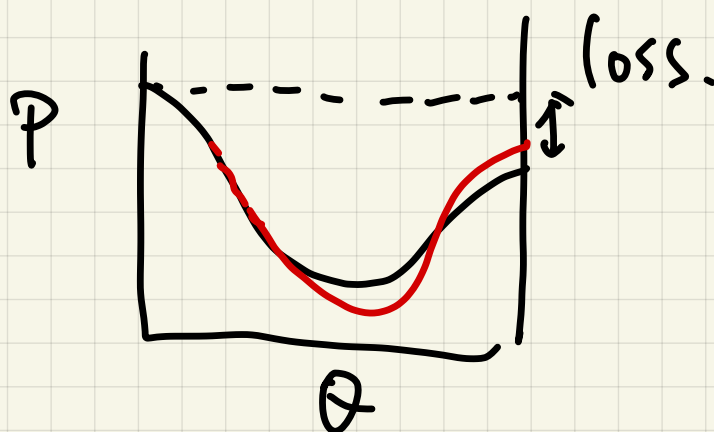
Golf ball
dimple



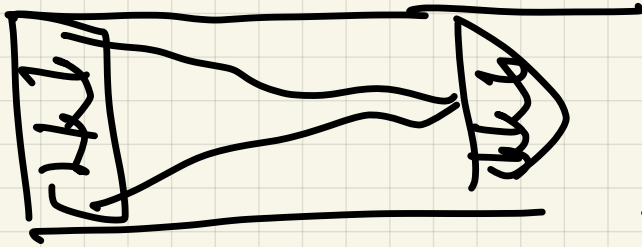
turbulence promotes mixing
and hence momentum
transport towards
the wall.

smaller drag

$$\Delta p < \Delta p$$



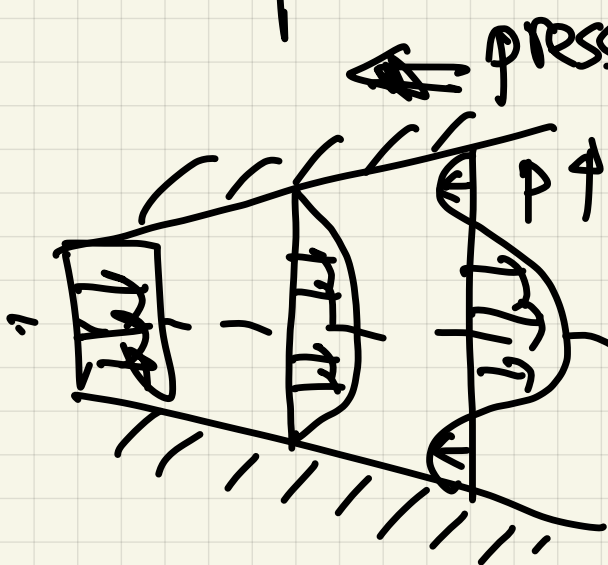
- Channel flow



$$Re = \frac{\rho U D}{\mu}$$

$Re > 2000$.
transition to
turbulence.

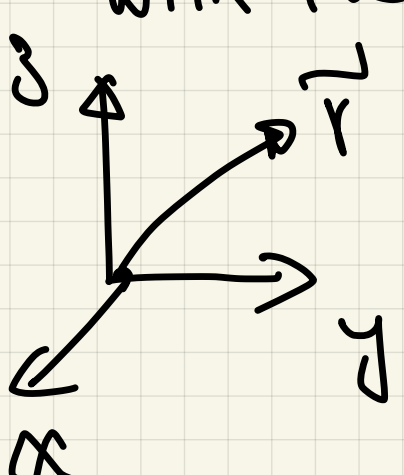
- Flow separation in a diffuser:



- lower momentum near the wall
- adverse pressure gradient
- reverse flow.
- turbulence tends to delay the separation.

- Kinematic properties

- For fluid mechanics, one concerns with the fluid velocity.


$$\vec{v}(\vec{r}, t) = \vec{v}(x, y, z, t).$$
$$= u(x, y, z, t)\vec{i} + v(x, y, z, t)\vec{j} + w(x, y, z, t)\vec{k}$$

- Fundamental laws of fluid mechanics

- conservation of mass, momentum and energy.

- Newton's 2nd law.

$$M\vec{a} = M \frac{D\vec{v}}{Dt} = \sum \vec{F}$$

- Lagrangian and Eulerian approaches.

$$Q(x, y, z, t).$$

$$dQ = \frac{\partial Q}{\partial x} dx + \frac{\partial Q}{\partial y} dy + \frac{\partial Q}{\partial z} dz + \frac{\partial Q}{\partial t} dt.$$

$$\frac{DQ}{Dt} = \frac{dQ}{dt} = \frac{\partial Q}{\partial t} + u \frac{\partial Q}{\partial x} + v \frac{\partial Q}{\partial y} + w \frac{\partial Q}{\partial z}$$

substantial
particle
material

derivative

local
derivative

convective
derivative

$\nabla \cdot \vec{v}$
vanish if

$$\vec{v} = 0$$

or

Q is homogeneous

$$\frac{DQ}{Dt}$$

Lagrangian

$$\frac{\partial Q}{\partial t} + (\vec{v} \cdot \nabla) Q$$

Eulerian

$$M \frac{D\vec{v}}{Dt} = \vec{F}$$

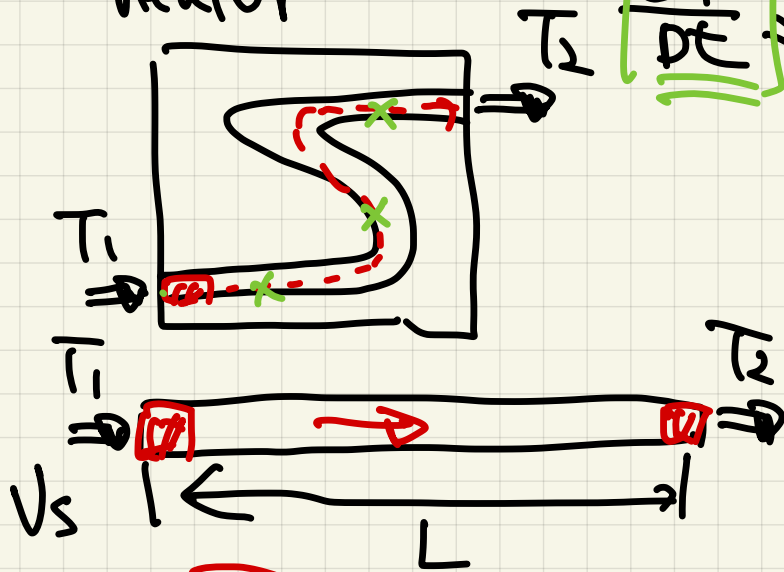
$$\frac{D\vec{v}}{Dt} = \frac{\vec{F}}{M}$$

Lagrangian

$$\frac{\partial \vec{v}}{\partial t} + \vec{v} \cdot \nabla \vec{v}$$

Eulerian

- Example heater



$$\Delta t = \frac{L}{V_s}$$

$$\frac{DT}{Dt} = \frac{T_2 - T_1}{\Delta t}$$

~~Lagrangian~~ Lagrangian

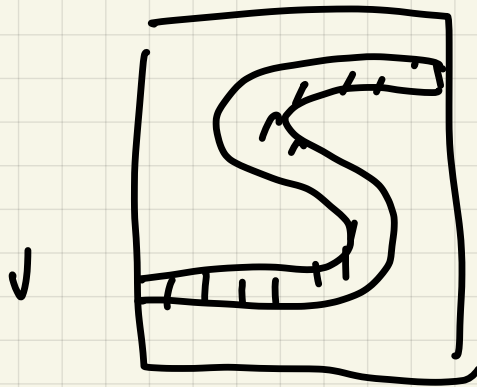
$$= \frac{dT}{dt} + V_s \frac{dT}{ds}$$

$$0 = V_s \frac{T_2 - T_1}{L/V_s}$$

~~Eulerian~~ Eulerian

$$= \frac{T_2 - T_1}{L/V_s}$$

- no velocity



$$\frac{DT}{Dt} = \frac{T(t+\Delta t) - T(t)}{\Delta t}$$

$$= \frac{dT}{dt} + V_s \frac{dT}{ds}$$

$$= \frac{T(t+\Delta t) - T(t)}{\Delta t}$$

$$\frac{DQ}{Dt} = \frac{\partial Q}{\partial t} + (\vec{v} \cdot \nabla) Q$$

$$\uparrow = \frac{\partial Q}{\partial t} + u \frac{\partial Q}{\partial x} + v \frac{\partial Q}{\partial y} + w \frac{\partial Q}{\partial z}$$

Lagrangian

Eulerian

• Acceleration of a particle

$$Q = \vec{v}$$

$$\frac{D\vec{v}}{Dt} = \frac{\partial \vec{v}}{\partial t} + \underbrace{(\vec{v} \cdot \nabla) \vec{v}}$$

vorticity

$$\nabla \left(\frac{\vec{v} \cdot \vec{v}}{2} \right) = \vec{v} \times (\nabla \times \vec{v})$$

$$\rho \frac{D\vec{v}}{Dt} + \rho (\vec{v} \cdot \nabla) \vec{v} = -\nabla p + \mu \nabla^2 \vec{v}$$

$$\rho \frac{D\vec{v}}{Dt} + \rho \nabla \left(\frac{\vec{v} \cdot \vec{v}}{2} \right) - \vec{v} \times (\nabla \times \vec{v}) = -\nabla p + \mu \nabla^2 \vec{v}$$

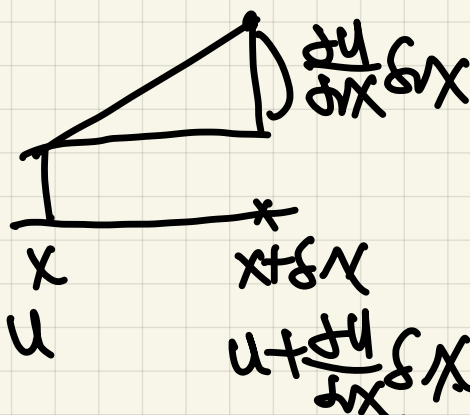
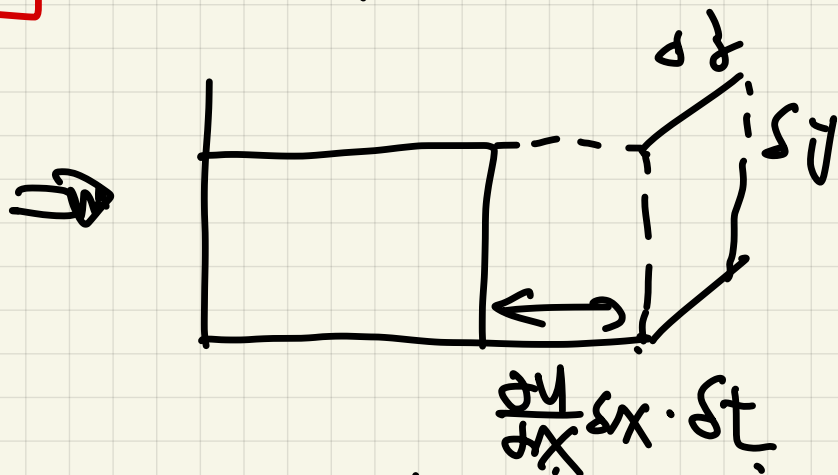
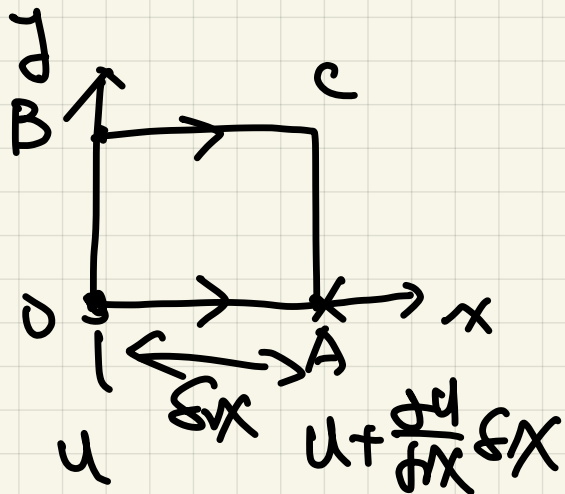
$$\rho \frac{D\vec{v}}{Dt} + \nabla \left(\frac{v^2}{2} + p \right) - \vec{v} \times (\nabla \times \vec{v}) = \mu \nabla^2 \vec{v}$$

2 for $\nabla \times \vec{v} \Rightarrow$ irrotational and $\mu \neq 0$.

③ Steady $\Rightarrow \frac{v^2}{2} + p = \text{const}$

$$Q = m \dot{q}, \quad \dot{q} = 1, \quad m = \rho \delta V$$

$$\frac{Dm}{Dt} = 0 = \rho \frac{D\delta V}{Dt} + \delta V \frac{D\rho}{Dt}$$



$$\delta V = \frac{\partial u}{\partial x} \delta x \delta t \cdot \delta y \delta z$$

change in control volume in δt .

— volume change rate per unit volume in x direction.

$$\frac{1}{\delta V} \frac{d\delta V}{dt} = \lim_{\delta t \rightarrow 0} \frac{\frac{\partial u}{\partial x} \delta t}{\delta t} = \frac{\partial u}{\partial x}$$

$$\Rightarrow \frac{1}{\delta V} \frac{d\delta V}{dt} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z}, \quad \nabla \cdot \mathbf{v}$$

$$\begin{aligned}
 \frac{Dm}{Dt} &\stackrel{!}{=} 0 = \rho \frac{D\delta V}{Dt} + \delta V \frac{D\rho}{Dt} \\
 &= \rho \delta V \nabla \cdot \vec{v} + \delta V \left[\frac{\partial \rho}{\partial t} + (\vec{v} \cdot \nabla) \rho \right] \\
 &= \delta V \left[\frac{\partial \rho}{\partial t} + \rho \nabla \cdot \vec{v} + (\vec{v} \cdot \nabla) \rho \right] \\
 &= \delta V \left[\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) \right] = 0
 \end{aligned}$$

- conservation of mass.

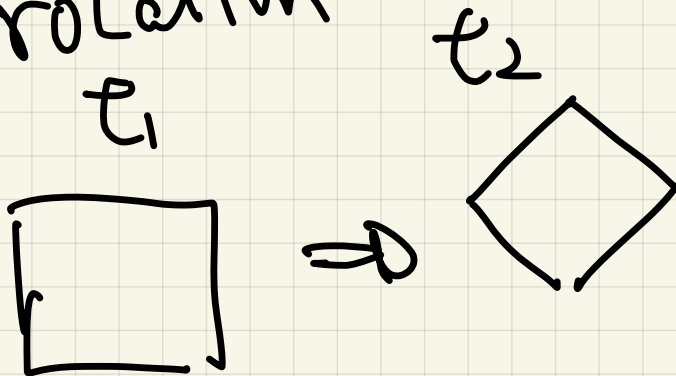
$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) = \frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x_j} (\rho v_j) = 0.$$

- Fluid element can undergo

• translation



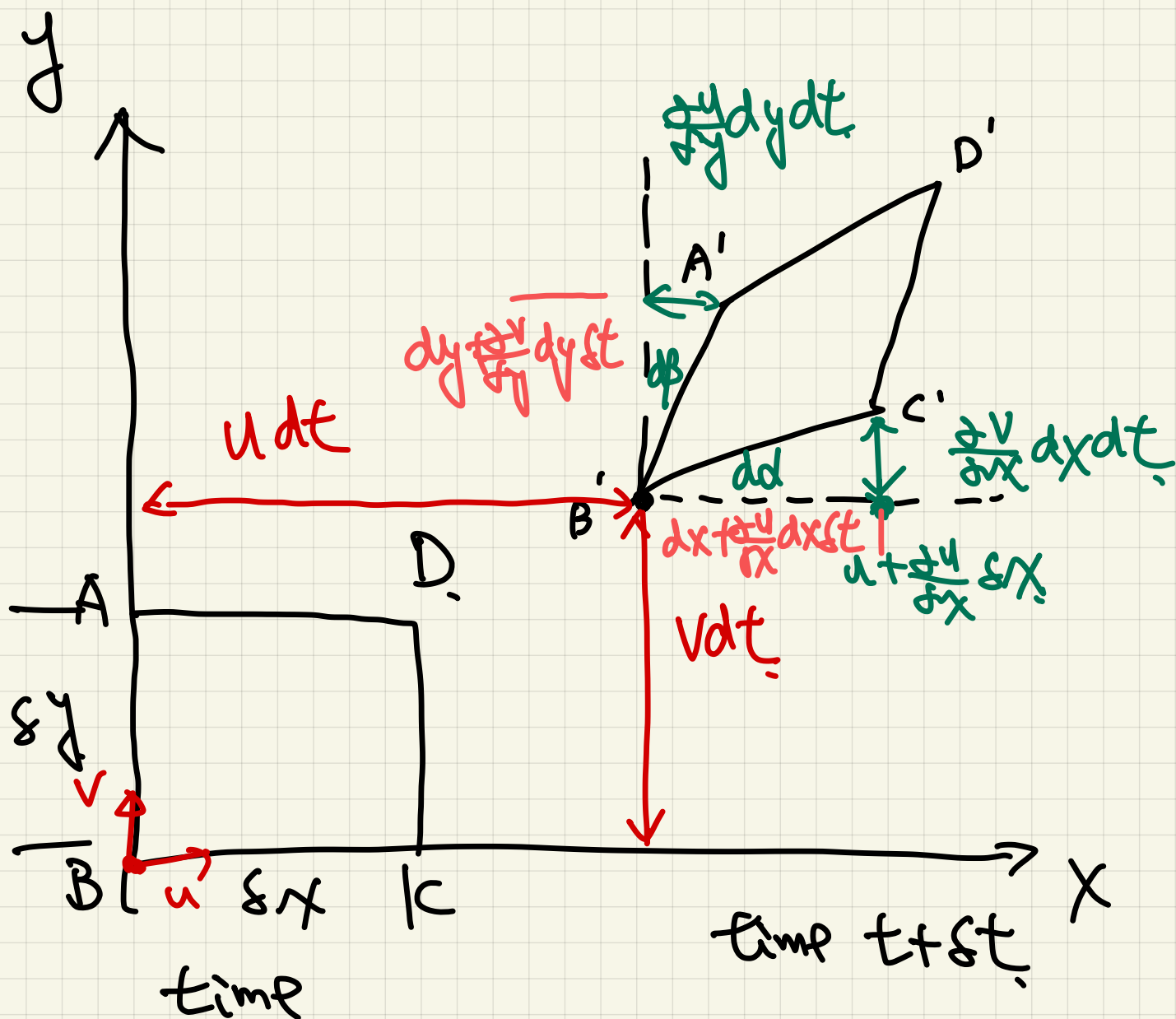
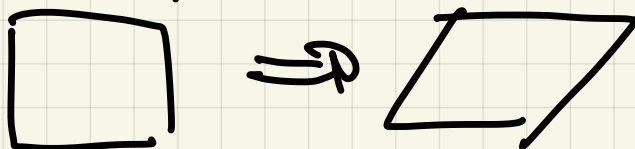
• rotation



• extensional strain.
 t_1 t_2



• shear
 t_1 t_2

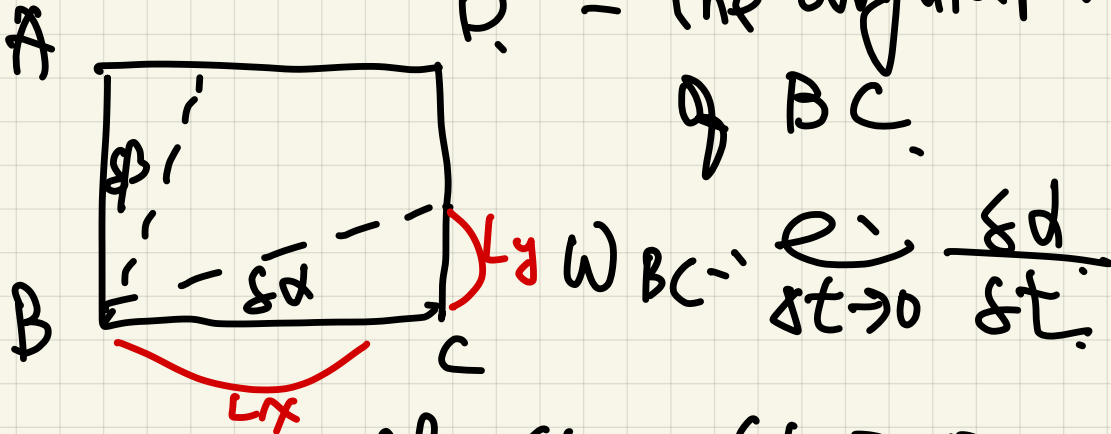


- the rate of translation

$$u = \frac{u \, \delta t}{\delta t}, \quad v = \frac{v \, \delta t}{\delta t}, \quad w = \frac{w \, \delta t}{\delta t}$$

- the rate of rotation

D. - the angular velocity of BC.



For small δt $\delta t \rightarrow 0$.

$$\tan \delta \alpha \approx \delta \alpha = \lim^{-1} \frac{\frac{\partial v}{\partial x} \delta x \, \delta t}{\delta x + \frac{\partial v}{\partial x} \delta x \, \delta t} = \frac{L_y}{L_x}$$

$$= \frac{\partial v}{\partial x} \delta t$$

$$\tan \delta \beta = \delta \beta = \lim^{-1} \frac{\frac{\partial v}{\partial y} \delta y \, \delta t}{\delta y + \frac{\partial v}{\partial y} \delta y \, \delta t} = \frac{\partial v}{\partial y} \delta t$$

D - The rotation of $\frac{dR_z}{dt}$ is the average of the angular velocity of

$\overline{BC}$ and $\overline{BA}$,

- If counter clockwise is positive.
then

$$\frac{dR_z}{dt} = \frac{1}{2} \left(\frac{dR_{Bx}}{dt} - \frac{dR_{BA}}{dt} \right) = \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right)$$

$$\frac{dR_x}{dt} = \frac{1}{2} \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \right), \quad \frac{dR_y}{dt} = \frac{1}{2} \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \right)$$

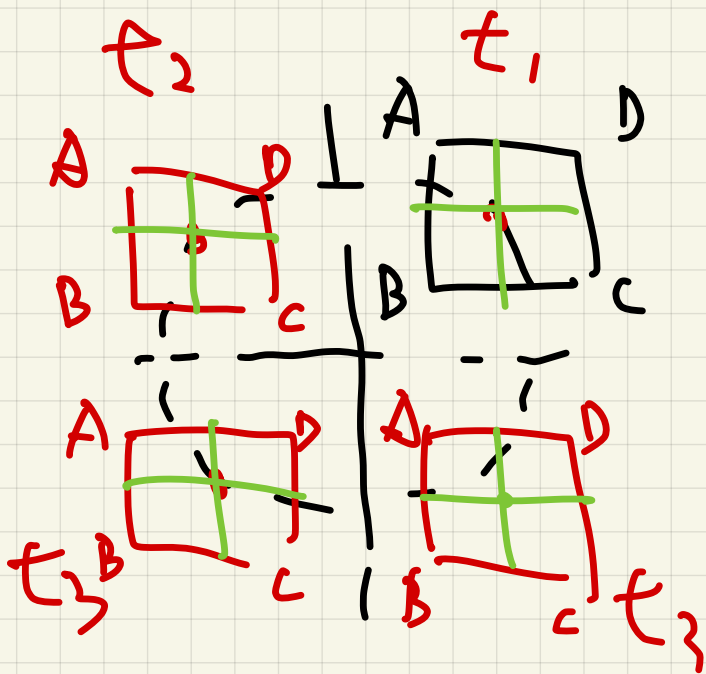
- Vorticity of a fluid

$$\vec{\omega} = 2 \frac{d\vec{R}}{dt} = \nabla \times \vec{v}$$

$$\omega_x = \frac{\partial}{\partial y} \left(\frac{\partial w}{\partial z} - \frac{\partial v}{\partial z} \right)$$

$$\omega_y = \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \right)$$

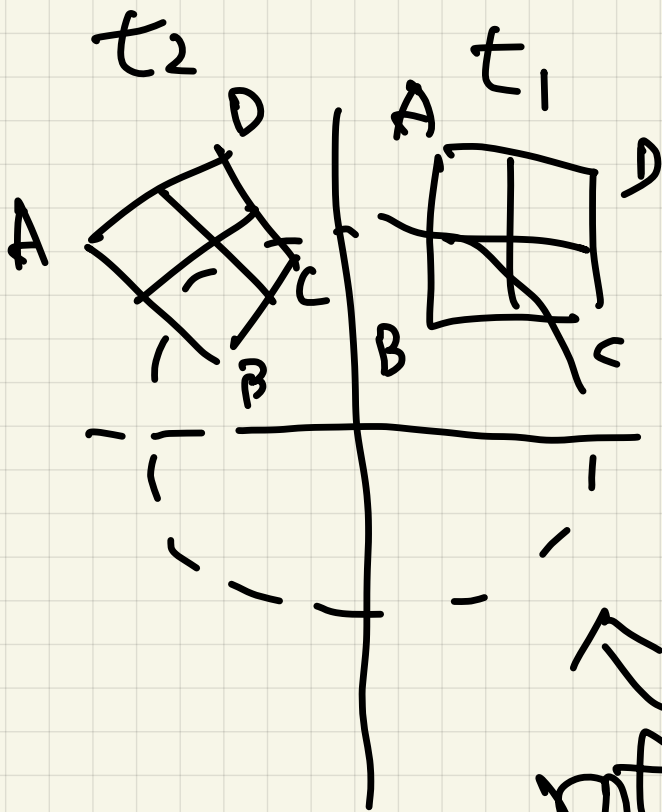
$$\omega_z = \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial y} - \frac{\partial v}{\partial x} \right)$$



non-rotational

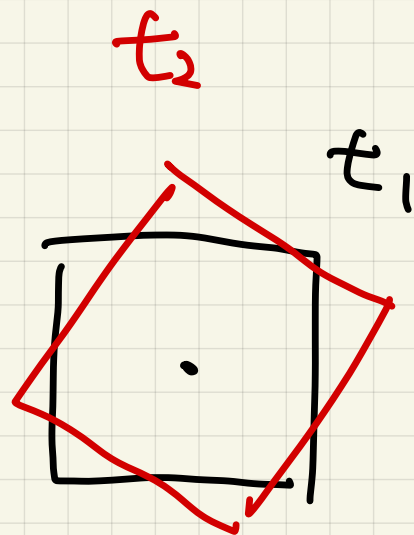
$$\vec{L} = 0.$$

$$\vec{V} \times \vec{V} = 0$$



rotational. $\vec{L} \neq 0.$

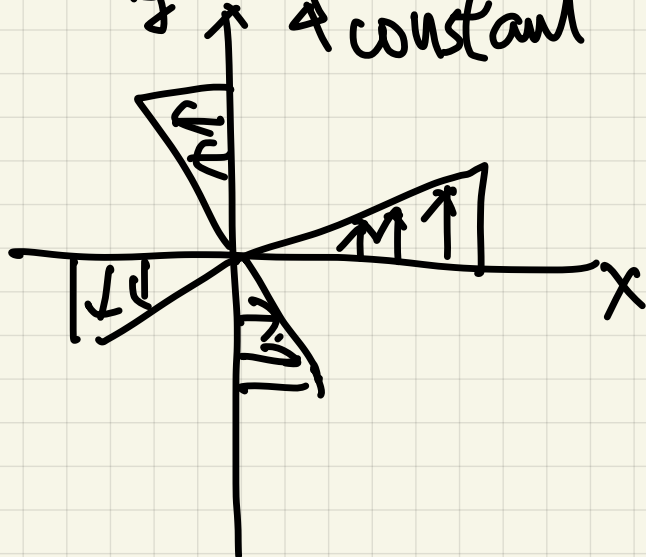
$$\vec{V} \times \vec{V} \neq 0$$



— Consider
Forced vortex

$$V_\theta = K r$$

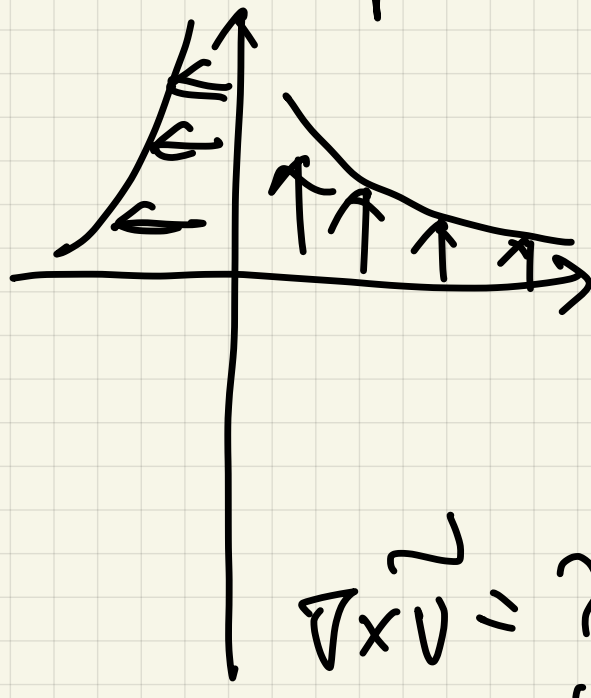
constant



$$\nabla \times \vec{V} = ? = \omega_z$$

free vortex bath tub

$$V_\theta = \frac{K}{r}$$



$$\nabla \times \vec{V} = ?$$

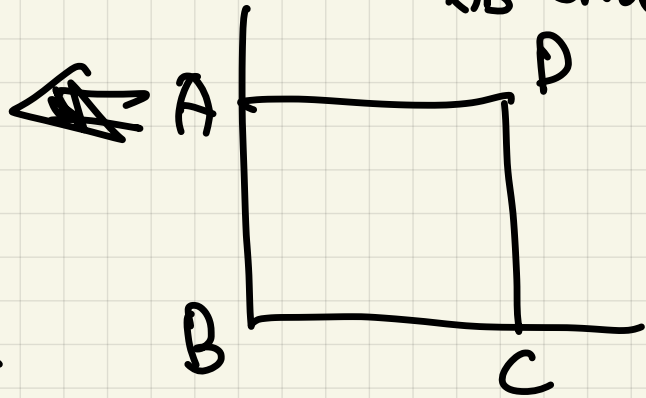
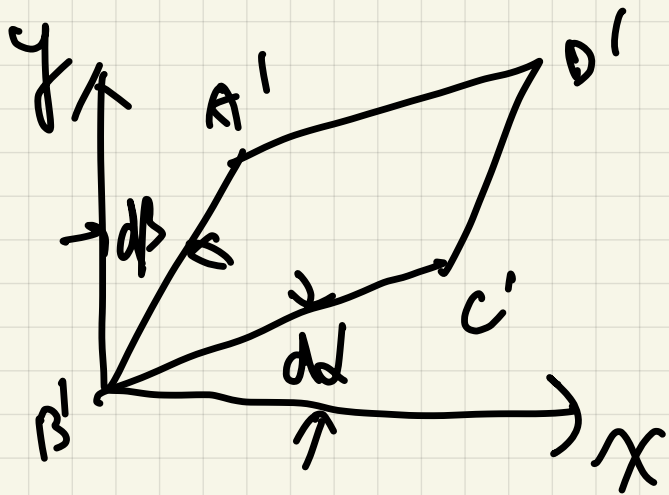
Take home.

— Determine forced vortex ~~is~~ rotational
free vortex ~~is~~ irrotational
rotational or irrotational

$$\omega_z = \frac{1}{r} \frac{d}{dr} (r V_\theta) - \frac{1}{r} \frac{dV_r}{dr}$$

$\frac{d(rKr)}{dr} \neq 0$
 $\frac{d(r \frac{K}{r})}{dr} = 0$

- Shear strain: the average decrease of the angle between $\overline{AB}$ and $\overline{BC}$.



$$\epsilon_{xy} = \frac{1}{2} \left(\frac{dx}{dy} + \frac{db}{dx} \right) = \frac{1}{2} \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right)$$

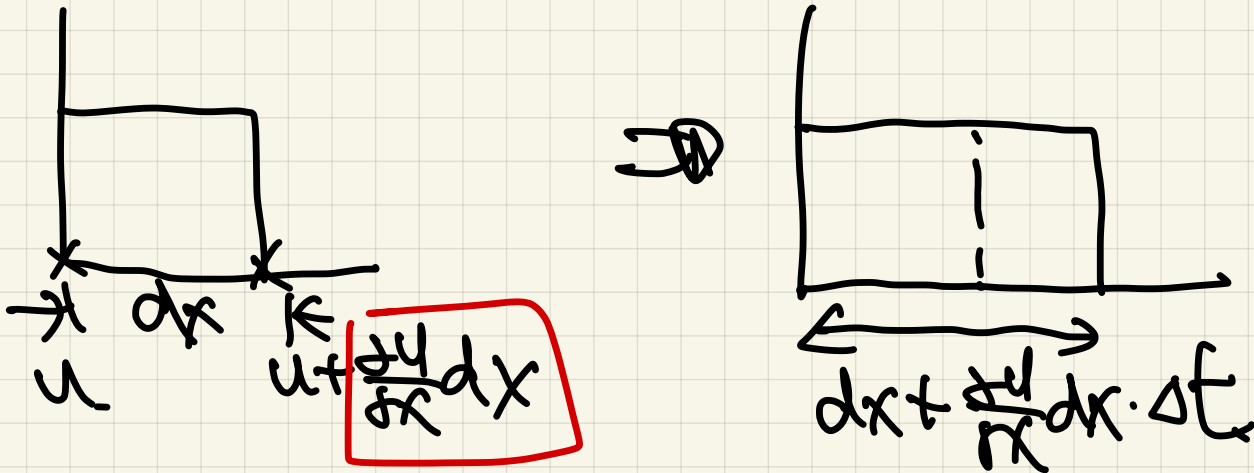
$$\epsilon_{yz} = \frac{1}{2} \left(\frac{\partial w}{\partial y} + \frac{\partial v}{\partial z} \right), \quad \epsilon_{zx} = \frac{1}{2} \left(\frac{\partial w}{\partial z} + \frac{\partial u}{\partial x} \right)$$

The shear strain rates are

symmetric $\epsilon_{ij} = \epsilon_{ji}$

$$\epsilon_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)$$

- Dilational or extensional strain

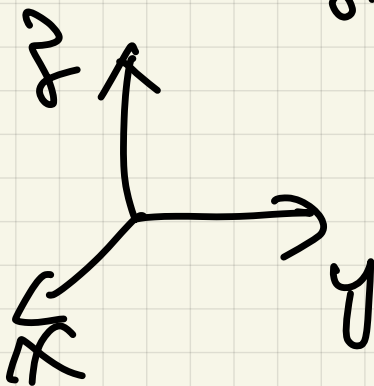


$$\Delta t \epsilon_{xx} = \frac{dx + \frac{du}{dx} dx \Delta t - dx}{dx} = \frac{du}{dx} \Delta t$$

$$\epsilon_{xx} = \frac{du}{dx}, \quad \epsilon_{yy} = \frac{dv}{dy}, \quad \epsilon_{zz} = \frac{dw}{dz}$$

- The strain rate tensor

$$\epsilon_{ij} = \begin{pmatrix} \epsilon_{xx} & \epsilon_{xy} & \epsilon_{xz} \\ \epsilon_{yx} & \epsilon_{yy} & \epsilon_{yz} \\ \epsilon_{zx} & \epsilon_{zy} & \epsilon_{zz} \end{pmatrix} \text{ symmetric and rank 2 tensor.}$$



- There are three invariants which are independent of direction or choice of axes.

$$I_1 = \epsilon_{xx} + \epsilon_{yy} + \epsilon_{zz} = \epsilon_1 + \epsilon_2 + \epsilon_3$$

$$I_2 = \epsilon_{xx}\epsilon_{yy} + \epsilon_{yy}\epsilon_{zz} + \epsilon_{zz}\epsilon_{xx} - \epsilon_{xy}^2 - \epsilon_{yx}^2 - \epsilon_{yz}^2 = \epsilon_1\epsilon_2 + \epsilon_2\epsilon_3 + \epsilon_3\epsilon_1$$

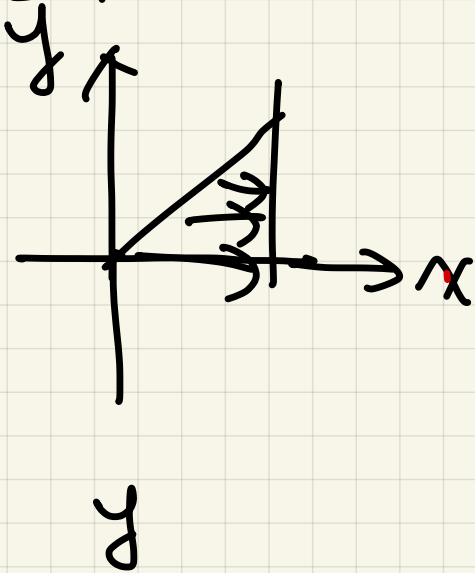
$$I_3 = \begin{vmatrix} \epsilon_{xx} & \epsilon_{xy} & \epsilon_{xz} \\ \epsilon_{yx} & \epsilon_{yy} & \epsilon_{yz} \\ \epsilon_{zx} & \epsilon_{zy} & \epsilon_{zz} \end{vmatrix} = \epsilon_1\epsilon_2\epsilon_3$$

- For symmetric tensor, there are principal axes, for which the strain rate tensors become,

$$\begin{vmatrix} \epsilon_1 & 0 & 0 \\ 0 & \epsilon_2 & 0 \\ 0 & 0 & \epsilon_3 \end{vmatrix} \quad \text{where } \epsilon_1, \epsilon_2, \epsilon_3 \text{ are called the principal strain rates}$$

If I_1, I_2, I_3 are known, $\epsilon_1, \epsilon_2, \epsilon_3$ can be obtained.

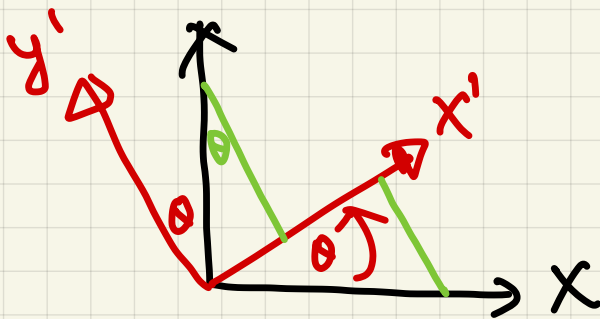
- Determination of the principal axes: - 2D



$$u = ay + b, \quad \frac{\partial u}{\partial x} = 0, \quad \frac{\partial u}{\partial y} = a$$

$$v = 0$$

$$\begin{bmatrix} \frac{\partial u}{\partial x}, \frac{1}{2} \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) \\ \frac{1}{2} \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right), \frac{\partial v}{\partial y} \end{bmatrix} = \begin{bmatrix} 0, \frac{1}{2}a \\ \frac{1}{2}a, 0 \end{bmatrix}$$



$$x' = x \cos \theta + y \sin \theta$$

$$y' = -x \sin \theta + y \cos \theta$$

$$u_{x'} = u_x \cos \theta + v_y \sin \theta$$

$$v_{y'} = -u_x \sin \theta + v_y \cos \theta$$

$$\begin{bmatrix} \frac{\partial u_{x'}}{\partial x'}, \frac{1}{2} \left(\frac{\partial u_{x'}}{\partial y'} + \frac{\partial v_{y'}}{\partial x'} \right) \\ \frac{1}{2} \left(\frac{\partial u_{x'}}{\partial y'} + \frac{\partial v_{y'}}{\partial x'} \right), \frac{\partial v_{y'}}{\partial y'} \end{bmatrix}$$

$$v_y > 0, \quad u_{x'} = u_x \cos \theta = (ay + b) \cos \theta$$

$$v_{y'} = -u_x \sin \theta = -(ay + b) \sin \theta$$

$$u_{x'} = [a(\underline{y' \cos \theta + x' \sin \theta}) + b] \cos \theta$$

$$v_{y'} = -[a(y' \cos \theta + x' \sin \theta) + b] \sin \theta$$

$$\frac{\partial u_{x'}}{\partial x'} = a \sin \theta \cos \theta, \quad \frac{\partial u_{x'}}{\partial y'} = a \cos \theta \cos \theta, \quad \frac{\partial v_{y'}}{\partial x'} = -a \sin \theta \sin \theta$$

$$\frac{\partial u}{\partial x'} = a \sin \theta \cos \theta,$$

$$\frac{\partial u}{\partial y'} = a \cos \theta \cos \theta,$$

$$\frac{\partial v}{\partial x'} = -a \sin \theta \sin \theta,$$

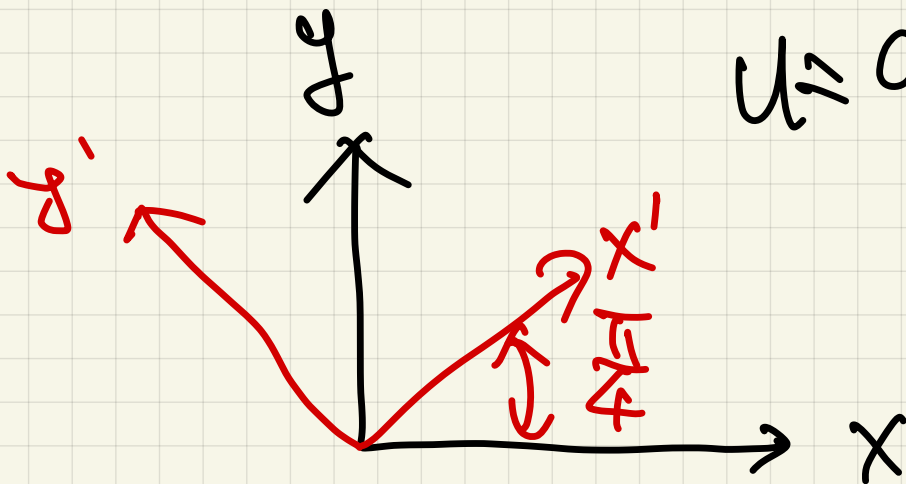
$$\frac{\partial v}{\partial y'} = -a \cos \theta \sin \theta$$

$$\begin{bmatrix} \frac{\partial u}{\partial x'}, & \frac{1}{2} \left(\frac{\partial u}{\partial y'} + \frac{\partial v}{\partial x'} \right) \\ \frac{1}{2} \left(\frac{\partial u}{\partial y'} + \frac{\partial v}{\partial x'} \right), & \frac{\partial v}{\partial y'} \end{bmatrix}$$

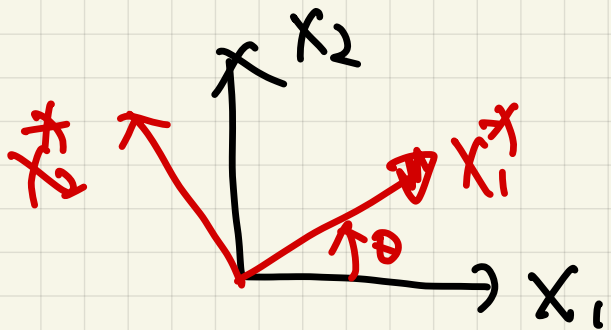
$$= \begin{bmatrix} a \sin \theta \cos \theta, & \frac{1}{2} a (\cos^2 \theta - \sin^2 \theta) \\ \frac{1}{2} a (\cos^2 \theta - \sin^2 \theta), & -a \sin \theta \cos \theta \end{bmatrix}$$

$$\cos^2 \theta - \sin^2 \theta = 0 \Rightarrow \theta = \frac{\pi}{4}$$

$$u = ay + b.$$



- coordinate transformation



$$x_1^* = x_1 \cos \theta + x_2 \sin \theta$$

$$x_2^* = -x_1 \sin \theta + x_2 \cos \theta$$

$$x_1^* = x_1 \cdot \vec{i}_1 \cdot \vec{i}_1^* + x_2 \cdot \vec{i}_2 \cdot \vec{i}_1^*$$

direction cosine

$$x_2^* = x_1 \cdot \vec{i}_1 \cdot \vec{i}_2^* + x_2 \cdot \vec{i}_2 \cdot \vec{i}_2^*$$

$$x_i^* = \lambda_{ij} x_j, \quad \lambda_{ij} = \vec{i}_i \cdot \vec{j} \quad \text{direction cosine}$$

$$= \lambda_{11} x_1 + \lambda_{12} x_2 + \lambda_{13} x_3$$

$$a_{mn}^* = \lambda_{mi} \lambda_{nj} \vec{i}_j, \quad \lambda_{mi} = \vec{i}_m \cdot \vec{i}_i$$

$$\vec{x}^* = \vec{Q} \vec{x}$$

$$\vec{Q} = \begin{bmatrix} \lambda_{11} & \lambda_{12} & \lambda_{13} \\ \lambda_{21} & \lambda_{22} & \lambda_{23} \\ \lambda_{31} & \lambda_{32} & \lambda_{33} \end{bmatrix}$$

$$\vec{x} = \vec{Q} \vec{x}^* \vec{Q}^T$$

$$\vec{Q}^T = \begin{bmatrix} \lambda_{11} & \lambda_{21} & \lambda_{31} \\ \lambda_{12} & \lambda_{22} & \lambda_{32} \\ \lambda_{13} & \lambda_{23} & \lambda_{33} \end{bmatrix}$$

$$X^* = \lambda_1 X_1 + \lambda_2 X_2 + \lambda_3 X_3$$

$$\begin{aligned} \rho_{mn}^* = & \lambda_{m1} \lambda_{n1} \rho_{11} + \lambda_{m1} \lambda_{n2} \rho_{12} + \lambda_{m1} \lambda_{n3} \rho_{13} \\ & + \lambda_{m2} \lambda_{n1} \rho_{21} + \lambda_{m2} \lambda_{n2} \rho_{22} + \lambda_{m2} \lambda_{n3} \rho_{23} \\ & + \lambda_{m3} \lambda_{n1} \rho_{31} + \lambda_{m3} \lambda_{n2} \rho_{32} + \lambda_{m3} \lambda_{n3} \rho_{33} \end{aligned}$$

$$\begin{bmatrix} \rho_{xx}^* & \rho_{xy}^* \\ \rho_{yx}^* & \rho_{yy}^* \end{bmatrix} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} \rho_{xx} & \rho_{xy} \\ \rho_{yx} & \rho_{yy} \end{bmatrix} \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

$$\rho_{xx}^* = \rho_{xx} \cos^2\theta + \rho_{yy} \sin^2\theta + 2\rho_{xy} \sin\theta \cos\theta$$

$$\rho_{yy}^* = \rho_{xx} \sin^2\theta + \rho_{yy} \cos^2\theta - 2\rho_{xy} \sin\theta \cos\theta$$

$$\rho_{xy}^* = (\rho_{yy} - \rho_{xx}) \sin\theta \cos\theta + \rho_{xy} (\cos^2\theta - \sin^2\theta)$$

$$\rho_{xy}^* = 0$$

Take home 2D find the principal axes.

$$V = ax + by \Rightarrow \theta = ?$$

$$\rho_{xy}^* = 0$$

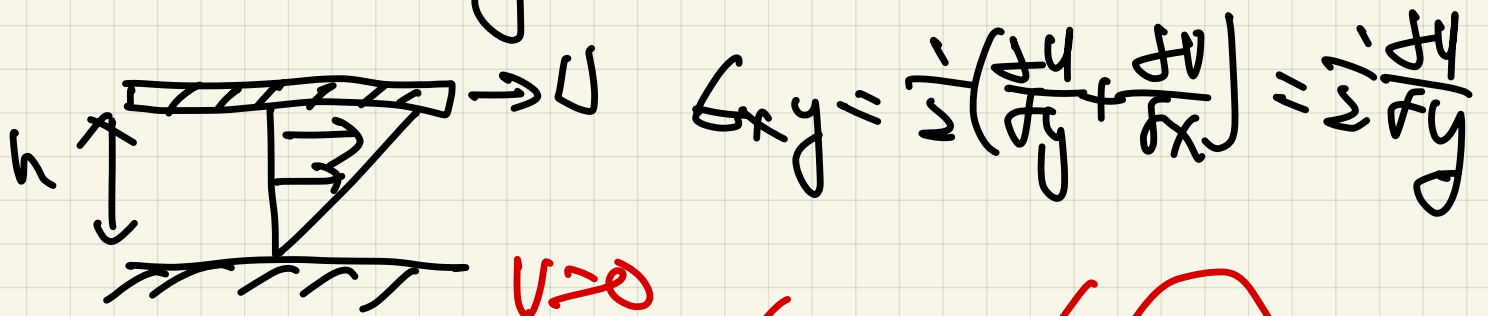
- symmetric and axi-symmetric strain rates.

$$\frac{\partial u_i}{\partial x_j} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) + \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} - \frac{\partial u_j}{\partial x_i} \right)$$

$= \epsilon_{ij} + \omega_{ij}$
strain rate. angular velocity

The angular velocity does not distort the elements, so that only the strain rate will cause the viscous stress.

- Viscosity: how to determine



$$\epsilon_{xy} = \frac{1}{2} \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) = \frac{1}{2} \frac{\partial u}{\partial y}$$

$$\cancel{\rho \frac{\partial u}{\partial x}} + \cancel{\rho \frac{\partial v}{\partial y}} = - \cancel{\frac{\partial p}{\partial x}} + \mu \left(\cancel{\frac{\partial^2 u}{\partial x^2}} + \frac{\partial^2 u}{\partial y^2} \right)$$

$$\frac{\partial^2}{\partial y^2} \left(\mu \frac{\partial u}{\partial y} \right) = 0 \Rightarrow \frac{\partial}{\partial y} (\tau) = 0.$$

τ assume to be newtonian

- For newtonian fluid

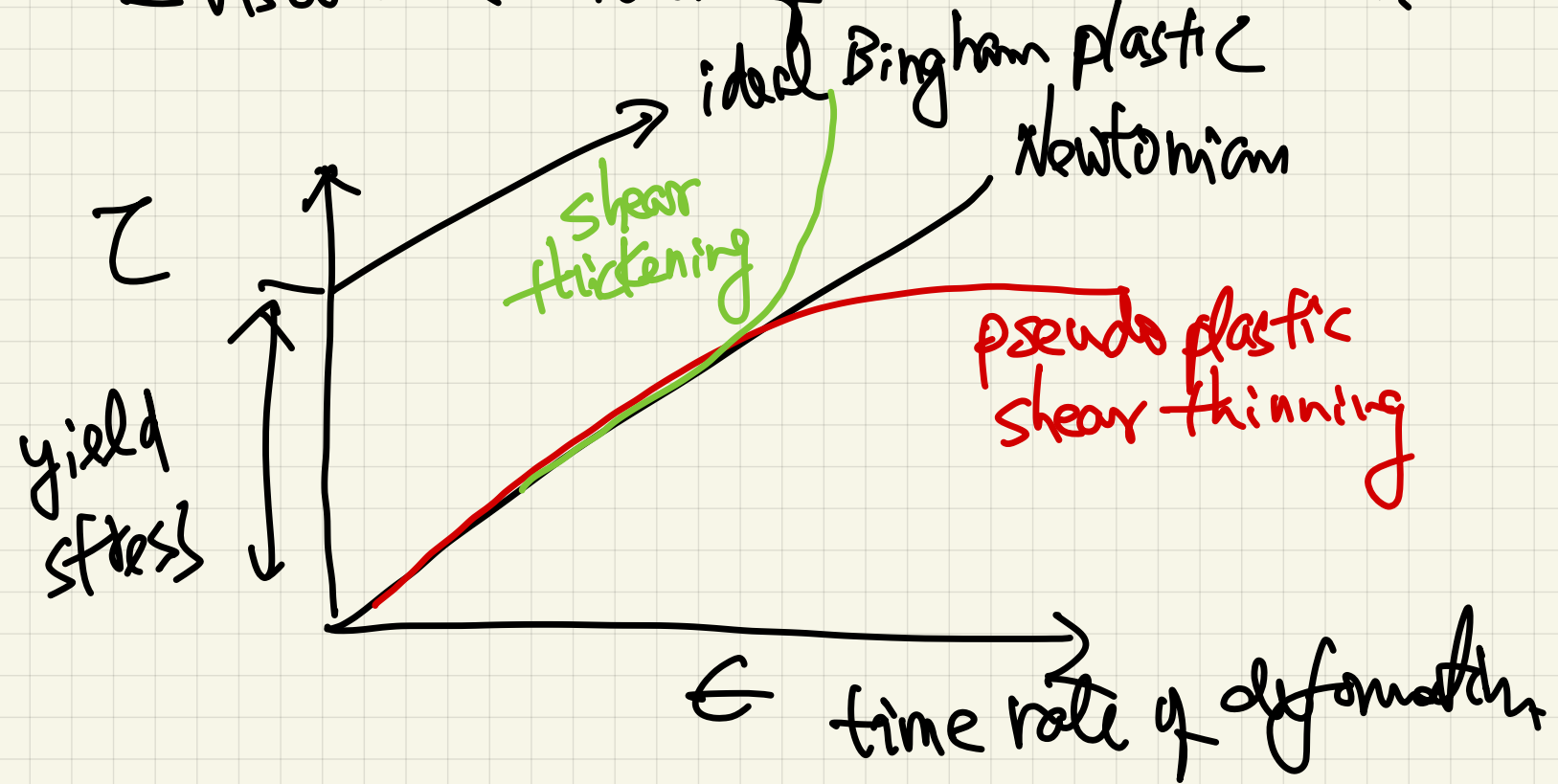
$$\tau_{xy} \propto \epsilon_{xy}, \quad \tau_{xy} = \mu \frac{U}{h} \Rightarrow \mu \epsilon_{xy} = \mu \frac{\partial u}{\partial y}$$

μ : coefficient of viscosity of a newtonian fluid.

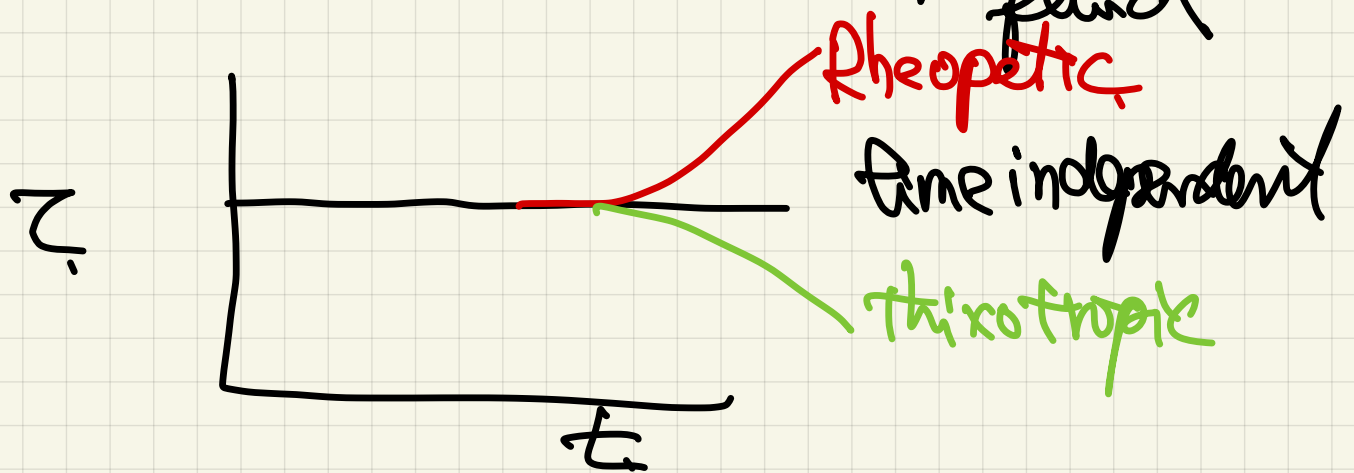
$$\tau_{ij} = \mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) + \delta_{ij} \lambda \frac{\partial u_k}{\partial x_k}$$

$$\lambda = -\frac{2}{3} \mu$$

— Viscous behaviour of various materials



— time dependent behaviour of non-newtonian fluid



- A simple approach for non-Newtonian behaviour

$$\tau_{xy} \sim \sigma_{xy}$$

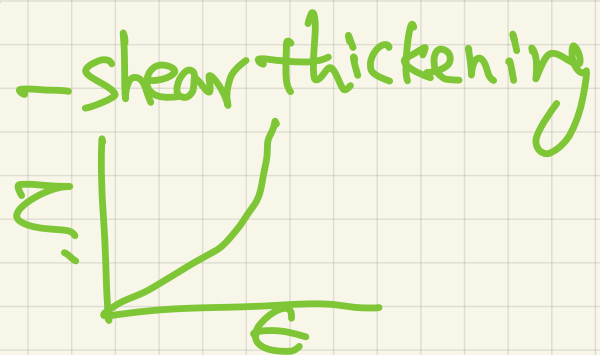
$$\tau_{xy} = \eta \sigma_{xy}$$

η unrealistic at the origin

$n < 1$ pseudo-plastic. - shear thinning

$n = 1$ Newtonian

$n > 1$ Dilatant



- The present study is confined to Newtonian flow

μ is function of T and P

1. μ of liquid $\downarrow$ as $T \uparrow$
2. μ of gas $\uparrow$ as $T \uparrow$
3. $\mu \uparrow$ as $P \uparrow$

$$\mu \approx 0.67 \rho \bar{c} \lambda \leftarrow \text{speed of sound}$$

density
mean free path

- For dilute gas.

$$\frac{\mu}{\mu_0} \approx \left(\frac{T}{T_0} \right)^n, \quad n=0.7.$$

T_0, μ_0 are reference values.

- formula by Sutherland.

$$\frac{\mu}{\mu_0} = \left(\frac{T}{T_0} \right)^{3/2} \frac{T_0 + S}{T + S}$$

S = Sutherland constant.